

Problem 1

Two bodies (A and B) are freely falling. Body B falls from a height of $h_B = 150$ m and lands earlier than body A by a time interval of $\Delta t = 3.5$ s. From what height was body A released?

Solution

The time taken for body B to fall is determined using the free-fall equation*:

$$h_B = \frac{1}{2}gt_B^2.$$

Solving for t_B :

$$t_B = \sqrt{\frac{2h_B}{g}} = \sqrt{\frac{2 \times 150}{9.81}} \approx \sqrt{30.58} \approx 5.53 \text{ s.}$$

Since body A takes $\Delta t = 3.5$ s longer to reach the ground, its total fall time is:

$$t_A = t_B + \Delta t = 5.53 + 3.5 = 9.03 \text{ s.}$$

Using the free-fall equation* for body A:

$$h_A = \frac{1}{2}gt_A^2.$$

Substituting t_A :

$$h_A = \frac{1}{2} \times 9.81 \times (9.03)^2.$$

Calculating:

$$h_A = 4.905 \times 81.54 \approx 399.85 \text{ m.}$$

Thus, body A was released from a height of approximately:

$$h_A \approx 400 \text{ m.}$$